

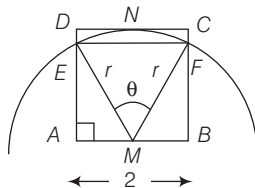
- 38.** The lengths of two sides of a right angled triangle which contain the right angle are a and b , respectively. Three squares are drawn on the three sides of the triangle on the outer side. What is the total area of the triangle and the three squares?

(a) $2(a^2 + b^2) + ab$ (b) $2(a^2 + b^2) + 2.5 ab$
(c) $2(a^2 + b^2) + 0.5 ab$ (d) $25(a^2 + b^2)$

- 39.** A grassy field has the shape of an equilateral triangle of side 6 m. A horse is tied to one of its vertices with a rope of length 4.2 m. The percentage of the total area of the field which is available for grazing is best approximated by

(a) 50% (b) 55% (c) 59% (d) 62%

- 40.** From the mid-point of the side of a square of length 2 units, a circle of radius 2 units is circumscribed. The area of intersection of the square and the circle is



(a) $\frac{4}{3}\pi + 3\sqrt{3}$ (b) $\pi + \sqrt{3}$ (c) $\frac{2}{3}\pi + \sqrt{3}$ (d) $\frac{2}{3}\pi + 2\sqrt{3}$

- 41.** In a circle with radius 2 cm, on its chord $2\sqrt{2}$ cm long taken as a diameter, another circle is constructed. The area of part of that circle, which is outside the greater circle is

(a) 1 cm^2 (b) 2 cm^2 (c) 6 cm^2 (d) 10 cm^2

- 42.** If a rectangular park has length l , width w , area a and perimeter p , which equation describing this park must be true?

I. $wp - 2w^2 = 2a$ II. $2a - 2w^2 - wp = 0$

III. $p^2 - 8a = 4(l^2 + w^2)$

Select the correct answer using the codes given below

(a) I and II (b) II and III
(c) I and III (d) All of these

- 43.** A regular polygon with number of sides 12 is inscribed in a circle of area $9\pi \text{ cm}^2$. If r is the radius of a circle inscribed in a polygon, then consider the following statements.

I. Radius of circumcircle is 3 cm.

II. Radius of incircle is $3\cos 15^\circ$.

III. Ratio of area of circumcircle to area of polygon is $\pi : 3$.

Which of the following statement(s) are correct?

(a) I and II (b) II and III
(c) I and III (d) All of these

- 44.** What is the area of rectangle R ?

I. The length of rectangle R is twice the width.

II. The area of rectangle R is twice the perimeter.

(a) Statement I alone is sufficient but statement II alone is not sufficient.

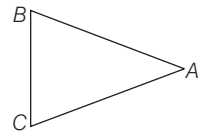
(b) Statement II alone is sufficient but statement I alone is not sufficient.

(c) Both statements together are sufficient.

(d) Neither statement I nor II is sufficient.

Directions (Q. Nos. 45-46) Answer the questions based on the following information.

A cow is tethered at point A by a rope. Neither the rope nor the cow is allowed to enter the triangle ABC . $\angle BAC = 30^\circ$, $AB = AC = 10 \text{ m}$.



- 45.** What is the area that can be grazed by the cow if the length of the rope is 8 m?

(a) $134\pi \frac{1}{3} \text{ m}^2$ (b) $121\pi \text{ m}^2$ (c) $132\pi \text{ m}^2$ (d) $\frac{176\pi}{3} \text{ m}^2$

- 46.** What is the area that can be grazed by the cow if the length of the rope is 12 m?

(a) $133\pi \frac{1}{6} \text{ m}^2$ (b) $121\pi \text{ m}^2$ (c) $132\pi \text{ m}^2$ (d) $\frac{176\pi}{3} \text{ m}^2$

PREVIOUS YEARS' QUESTIONS

- 47.** If the outer and inner diameters of a stone parapet around a well are 112 cm and 70 cm, respectively. Then, what is the area of the parapet? ☑ 2012 I

(a) 264 cm^2 (b) 3003 cm^2
(c) 6006 cm^2 (d) 24024 cm^2

- 48.** The area of a rectangle, whose one side is a is $2a^2$. What is the area of a square having one of the diagonals of the rectangle as side? ☑ 2012 I

(a) $2a^2$ (b) $3a^2$ (c) $4a^2$ (d) $5a^2$

- 49.** If the altitude of an equilateral triangle is $\sqrt{3} \text{ cm}$, then what is its perimeter? ☑ 2012 I

(a) 3 cm (b) $3\sqrt{3} \text{ cm}$ (c) 6 cm (d) $6\sqrt{3} \text{ cm}$

- 50.** If the area of a rectangle whose length is 5 more than twice its width is 75 sq units. What is the perimeter of the rectangle? ☑ 2012 I

(a) 40 units (b) 30 units
(c) 24 units (d) 20 units

- 51.** A square, a circle and an equilateral triangle have same perimeter.

Consider the following statements

I. The area of square is greater than the area of the triangle.

II. The area of circle is less than the area of triangle.

Which of the statement(s) given above is/are correct? **☑ 2012 I**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 52.** If the area of a circle is equal to the area of a square with side $2\sqrt{\pi}$ units, then what is the diameter of the circle? **☑ 2012 I**

- (a) 1 unit (b) 2 units (c) 4 units (d) 8 units

- 53.** In the $\triangle ABC$, the base BC is trisected at D and E . The line through D , parallel to AB , meets AC at F and the line through E parallel to AC meets AB at G . If EG and DF intersect at H , then what is the ratio of the sum of the area of parallelogram $AGHF$ and the area of the $\triangle DHE$ to the area of the $\triangle ABC$? **☑ 2012 II**

- (a) $1/2$ (b) $1/3$ (c) $1/4$ (d) $1/6$

- 54.** If the area of a circle inscribed in an equilateral triangle is 154 cm^2 , then what is the perimeter of the triangle? **☑ 2012 II**

- (a) 21 cm (b) $42\sqrt{3}$ cm (c) $21\sqrt{3}$ cm (d) 42 cm

- 55.** If the circumferences of two circles are in the ratio $2 : 3$, then what is the ratio of their areas? **☑ 2012 II**

- (a) $2 : 3$ (b) $4 : 9$ (c) $1 : 3$ (d) $8 : 27$

- 56.** If the circumference of a circle is equal to the perimeter of square, then which one of the following is correct? **☑ 2012 II**

- (a) Area of circle = Area of square
(b) Area of circle $\geq$ Area of square
(c) Area of circle $>$ Area of square
(d) Area of circle $<$ Area of square

- 57.** The area enclosed between the circumferences of two concentric circles is $16\pi \text{ cm}^2$ and their radii are in the ratio $5 : 3$. What is the area of the outer circle? **☑ 2012 II**

- (a) $9\pi \text{ cm}^2$ (b) $16\pi \text{ cm}^2$ (c) $25\pi \text{ cm}^2$ (d) $36\pi \text{ cm}^2$

- 58.** The perimeter of a rectangle is 82 m and its area is 400 m^2 . What is the breadth of the rectangle? **☑ 2012 II**

- (a) 18 m (b) 16 m (c) 14 m (d) 12 m

- 59.** Consider the following statements

- I. Area of a segment of a circle is less than area of its corresponding sector.
II. Distance travelled by a circular wheel of diameter $2d$ cm in one revolution is greater than $6d$ cm.

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II **☑ 2012 II**
(c) Both I and II (d) Neither I nor II

- 60.** The minute hand of a watch is 2.5 cm long. The distance its extreme end transverses in 40 min is

- (a) $10\pi/3$ cm (b) $3\pi/10$ cm **☑ 2013 I**
(c) $10/3$ cm (d) 10 cm

- 61.** The short and long hands of a clock are 4 cm and 6 cm long, respectively. Then, the ratio of distances travelled by tips of short hand in 2 days and long hand in 3 days is **☑ 2013 I**

- (a) $4 : 9$ (b) $2 : 9$ (c) $2 : 3$ (d) $1 : 27$

- 62.** The area of a square inscribed in a circle of radius 8 cm is **☑ 2013 I**

- (a) 32 cm^2 (b) 64 cm^2 (c) 128 cm^2 (d) 256 cm^2

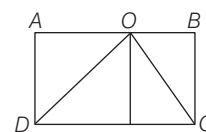
- 63.** A circular water fountain 6.6 m in diameter is surrounded by a path of width 1.5 m. The area of this path (in m^2) is **☑ 2013 I**

- (a) 13.62π (b) 13.15π
(c) 12.15π (d) None of these

- 64.** If an isosceles right angled triangle has area 1 sq unit, then what is its perimeter? **☑ 2013 I**

- (a) 3 units (b) $2\sqrt{2} + 1$ units
(c) $(\sqrt{2} + 1)$ units (d) $2(\sqrt{2} + 1)$ units

- 65.** In the figure given below, the area of rectangle $ABCD$ is 100 cm^2 , O is any point on AB and $CD = 20$ cm. Then, the area of $\triangle COD$ is **☑ 2013 I**



- (a) 40 cm^2 (b) 45 cm^2 (c) 50 cm^2 (d) 80 cm^2

- 66.** One side of a parallelogram is 8.06 cm and its perpendicular distance from opposite side is 2.08 cm. What is the approximate area of the parallelogram? **☑ 2013 II**

- (a) 12.56 cm^2 (b) 14.56 cm^2 (c) 16.76 cm^2 (d) 22.56 cm^2

- 67.** The perimeter of a rectangle having area equal to 144 cm^2 and sides in the ratio $4 : 9$ is **☑ 2013 II**

- (a) 52 cm (b) 56 cm (c) 60 cm (d) 64 cm

- 68.** What is the area between a square of side 10 cm and two inverted semi-circular, cross-section each of radius 5 cm inscribed in the square?

- (a) 17.5 cm^2 (b) 18.5 cm^2 **☑ 2013 II**
(c) 20.5 cm^2 (d) 21.5 cm^2

- 69.** How many 200 mm lengths can be cut from 10 m of ribbon? **☑ 2013 II**

- (a) 50 (b) 40 (c) 30 (d) 20

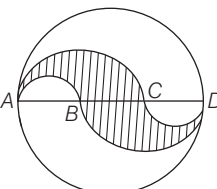
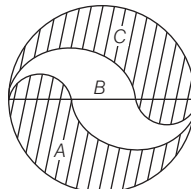
- 70.** The area of an isosceles $\triangle ABC$ with $AB = AC$ and altitude $AD = 3$ cm is 12 cm^2 . What is its perimeter? **☑ 2013 II**

- (a) 18 cm (b) 16 cm (c) 14 cm (d) 12 cm

- 71.** If the area of an equilateral triangle is x and its perimeter is y , then which one of the following is correct? **☑ 2013 II**

- (a) $y^4 = 432x^2$ (b) $y^4 = 216x^2$
(c) $y^2 = 432x^2$ (d) None of these

- 72.** What is the area of a circle whose area is equal to that of a triangle with sides 7 cm, 24 cm and 25 cm? ✎ 2013 II
 (a) 80 cm^2 (b) 84 cm^2 (c) 88 cm^2 (d) 90 cm^2
- 73.** If AB and CD are two diameters of a circle of radius r and they are mutually perpendicular, then what is the ratio of the area of the circle to the area of the ΔACD ? ✎ 2014 I
 (a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{4}$ (d) 2π
- 74.** A rectangle of maximum area is drawn inside a circle of diameter 5 cm. What is the maximum area of such a rectangle? ✎ 2014 I
 (a) 25 cm^2 (b) 12.5 cm^2
 (c) 12 cm^2 (d) None of these
- 75.** What is the area of the larger segment of a circle formed by a chord of length 5 cm subtending an angle of 90° at the centre? ✎ 2014 I
 (a) $\frac{25}{4}\left(\frac{\pi}{2} + 1\right)\text{ cm}^2$ (b) $\frac{25}{4}\left(\frac{\pi}{2} - 1\right)\text{ cm}^2$
 (c) $\frac{25}{4}\left(\frac{3\pi}{2} + 1\right)\text{ cm}^2$ (d) None of these
- 76.** Consider an equilateral triangle of side of unit length. A new equilateral triangle is formed by joining the mid-points of one, then a third equilateral triangle is formed by joining the mid-points of second. The process is continued. The perimeter of all triangles, thus formed is ✎ 2014 I
 (a) 2 units (b) 3 units (c) 6 units (d) Infinity
- 77.** ABC is triangle right angled at A , $AB = 6\text{ cm}$ and $AC = 8\text{ cm}$. Semi-circles drawn (outside the triangle) on AB , AC and BC as diameters which enclose areas x , y and z square units, respectively. What is $x + y - z$ equal to? ✎ 2014 I
 (a) 48 cm^2 (b) 32 cm^2
 (c) 0 (d) None of these
- 78.** A square is inscribed in a circle of diameter $2a$ and another square is circumscribing circle. The difference between the areas of outer and inner squares is ✎ 2014 I
 (a) a^2 (b) $2a^2$ (c) $3a^2$ (d) $4a^2$
- 79.** The area of a sector of a circle of radius 36 cm is $72\pi\text{ cm}^2$ the length of the corresponding arc of the sector is ✎ 2014 I
 (a) $\pi\text{ cm}$ (b) $3\pi\text{ cm}$ (c) $3\pi\text{ cm}$ (d) $4\pi\text{ cm}$
- 80.** What is the total area of three equilateral triangles inscribed in a semi-circle of radius 2 cm? ✎ 2014 I
 (a) 12 cm^2 (b) $\frac{3\sqrt{3}}{4}\text{ cm}^2$ (c) $\frac{9\sqrt{3}}{4}\text{ cm}^2$ (d) $3\sqrt{3}\text{ cm}^2$
- 81.** How many circular plates of diameter d be taken out of a square plate of side $2d$ with minimum loss of material? ✎ 2014 I
 (a) 8 (b) 6 (c) 4 (d) 2
- 82.** The sides of a triangle are in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$. If its perimeter is 52 cm, then what is the length of the smallest side? ✎ 2014 II
 (a) 9 cm (b) 10 cm (c) 11 cm (d) 12 cm
- 83.** The sides of a triangle are 25 cm, 39 cm and 56 cm. The perpendicular from the opposite vertex on the side of 56 cm is ✎ 2014 II
 (a) 10 cm (b) 12 cm (c) 15 cm (d) 16 cm
- 84.** The area of the largest triangle that can be inscribed in a semi-circle of radius r is ✎ 2015 I
 (a) r^2 (b) $2r^2$ (c) $3r^2$ (d) $4r^2$
- 85.** The area of a rhombus with side 13 cm and one diagonal 10 cm will be ✎ 2015 I
 (a) 140 cm^2 (b) 130 cm^2 (c) 120 cm^2 (d) 110 cm^2
- 86.** Four equal-sized maximum circular plates are cut off from a square paper sheet of area 784 cm^2 . The circumference of each plate is ✎ 2015 I
 (a) 11 cm (b) 22 cm (c) 33 cm (d) 44 cm
- 87.** From a circular piece of cardboard of radius 3 cm, two sectors of 40° each have been cut off. The area of the remaining portion is ✎ 2015 I
 (a) 11 cm^2 (b) 22 cm^2 (c) 33 cm^2 (d) 44 cm^2
- 88.** Three equal circles each of diameter d are drawn on a plane in such a way that each circle touches the other two circles. A big circle is drawn in such a manner that it touches each of the small circles internally. The area of the big circle is ✎ 2015 I
 (a) πd^2 (b) $\pi d^2(2 - \sqrt{3})^2$
 (c) $\frac{\pi d^2(\sqrt{3} + 1)^2}{2}$ (d) $\frac{\pi d^2(\sqrt{3} + 2)^2}{12}$
- 89.** The angles of a triangle are in the ratio 4 : 1 : 1. Then, the ratio of the largest side to the perimeter is ✎ 2015 I
 (a) $\frac{2}{3}$ (b) $\frac{1}{2 + \sqrt{3}}$ (c) $\frac{\sqrt{3}}{2 + \sqrt{3}}$ (d) $\frac{2}{1 + \sqrt{3}}$
- 90.** Let a , b , c be the sides of a right triangle, where c is the hypotenuse. The radius of the circle which touches the sides of the triangle is ✎ 2015 I
 (a) $(a + b - c)/2$ (b) $(a + b + c)/2$
 (c) $(a + 2b + 2c)/2$ (d) $(2a + 2b - c)/2$
- 91.** The ratio of the outer and inner perimeters of a circular path is 23 : 22. If the path is 5 m wide, the diameter of the inner circle is ✎ 2015 I
 (a) 55 m (b) 110 m (c) 220 m (d) 230 m

- 92.** The sides of a triangular field are 41 m, 40 m and 9 m. The number of rose beds that can be prepared in the field if each rose bed, on an average, needs 900 cm^2 space, is **2015 I**
 (a) 2000 (b) 1800 (c) 900 (d) 800
- 93.** The diameter of a wheel that makes 452 revolutions to move 2 km and 26 dm is equal to **2015 II**
 (a) $1\frac{9}{22}$ m (b) $1\frac{13}{22}$ m (c) $1\frac{5}{11}$ m (d) $1\frac{7}{11}$ m
- 94.** From a rectangular sheet of sides 18 cm and 14 cm, a semi-circular portion with smaller side as diameter is taken out. Then, the area of the remaining sheet will be **2015 II**
 (a) 98 cm^2 (b) 100 cm^2 (c) 108 cm^2 (d) 175 cm^2
- 95.** A square and an equilateral triangle have equal perimeter. If the diagonal of the square is $12\sqrt{2}$ cm, then the area of the triangle is **2015 II**
 (a) $24\sqrt{2} \text{ cm}^2$ (b) $24\sqrt{3} \text{ cm}^2$ (c) $48\sqrt{3} \text{ cm}^2$ (d) $64\sqrt{3} \text{ cm}^2$
- 96.** ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by **2015 II**
 (a) 50% (b) 52% (c) 54% (d) 56%
- 97.** The area of a trapezium is 336 cm^2 . If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is **2015 II**
 (a) 20 cm (b) 22 cm (c) 24 cm (d) 26 cm
- 98.** The circumference of a circle is 100 cm. The side of the square inscribed in the circle is **2015 II**
 (a) $50\sqrt{2} \text{ cm}$ (b) $\frac{100}{\pi} \text{ cm}$ (c) $\frac{50\sqrt{2}}{\pi} \text{ cm}$ (d) $\frac{100\sqrt{2}}{\pi} \text{ cm}$
- 99.** There are 437 fruit plants in an orchard planted in rows. The distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants. There is 1 m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of ₹ 100 per metre? **2015 II**
 (a) ₹ 15600 (b) ₹ 16800 (c) ₹ 18200
 (d) More information is required
- 100.** A square is inscribed in a right triangle with legs x and y and has common right angle with the triangle. The perimeter of the square is given by **2015 II**
 (a) $\frac{2xy}{x+y}$ (b) $\frac{4xy}{x+y}$ (c) $\frac{2xy}{\sqrt{x^2+y^2}}$ (d) $\frac{4xy}{\sqrt{x^2+y^2}}$
- 101.** A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p then the area of the polygon is **2015 II**
 (a) $(p+n)r$ (b) $(2p-n)r$ (c) $\frac{pr}{2}$ (d) None of these
- 102.** A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times the area of the smaller ring, then n is equal to **2016 I**
 (a) 4 (b) 6 (c) 8 (d) 10
- 103.** The number of rounds that a wheel of diameter $7/11$ m will make in traversing 4 km will be **2016 I**
 (a) 500 (b) 1000 (c) 1700 (d) 2000
- 104.** The base of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then, the area of the triangle is **2016 I**
 (a) 9600 sq units (b) 10000 sq units
 (c) 12000 sq units (d) None of these
- 105.** Four equal disc are placed such that each one touches two others. If the area of empty space enclosed by them is $150/847 \text{ cm}^2$, then the radius of each disc is equal to **2016 I**
 (a) $7/6 \text{ cm}$ (b) $5/6 \text{ cm}$ (c) $1/2 \text{ cm}$ (d) $5/11 \text{ cm}$
- 106.** AD is the diameter of a circle with area 707 m^2 and $AB = BC = CD$ as shown in the figure above. All curves inside the circle are semicircles with their diameters on AD. What is the cost of levelling the shaded region at the rate of ₹ 63 per m^2 ? **2016 I**
 (a) ₹ 29700 (b) ₹ 22400 (c) ₹ 14847 (d) None of these
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- 107.** A rhombus is formed by joining mid-points of the sides of a rectangle in the suitable order. If the area of the rhombus is 2 sq units, then the area of the rectangle is **2016 I**
 (a) $2\sqrt{2}$ sq units (b) 4 sq units
 (c) $4\sqrt{2}$ sq units (d) 8 sq units
- 108.** A circle of 3 m radius is divided into three areas by semi-circles of radii 1 m and 2 m as shown in the figure above. The ratio of the three areas A, B and C will be **2016 I**
 (a) 2 : 3 : 2 (b) 1 : 1 : 1 (c) 4 : 3 : 4 (d) 1 : 2 : 1
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ANSWERS

1	a	2	c	3	a	4	d	5	b	6	c	7	c	8	d	9	c	10	a
11	b	12	c	13	c	14	a	15	b	16	a	17	c	18	a	19	c	20	d
21	a	22	b	23	c	24	a	25	d	26	b	27	d	28	c	29	c	30	b
31	a	32	c	33	a	34	c	35	b	36	b	37	d	38	c	39	c	40	c
41	b	42	c	43	d	44	c	45	d	46	c	47	c	48	d	49	c	50	a
51	a	52	c	53	b	54	b	55	b	56	c	57	c	58	b	59	c	60	a
61	d	62	c	63	c	64	d	65	c	66	c	67	a	68	d	69	a	70	a
71	a	72	b	73	b	74	c	75	c	76	c	77	c	78	b	79	d	80	d
81	c	82	d	83	c	84	a	85	c	86	d	87	b	88	d	89	c	90	a
91	c	92	a	93	b	94	d	95	d	96	d	97	a	98	c	99	b	100	b
101	c	102	c	103	d	104	c	105	d	106	c	107	b	108	b				

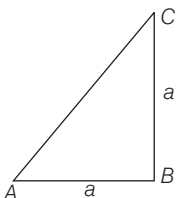
HINTS AND SOLUTIONS

1. (a) Area of square = $\frac{1}{2} \times (\text{Diagonal})^2$
 $= \frac{1}{2} \times 50 \times 50$
 $= 1250 \text{ m}^2$

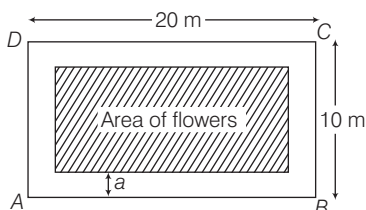
2. (c) Let length of rectangle be x cm.
 and breadth of rectangle = $(x - 2)$ cm
 $\therefore$ Perimeter of rectangle = $2(l + b)$
 So, $2[x + (x - 2)] = 48$
 $\Rightarrow 4x - 4 = 48$
 $\Rightarrow x = \frac{52}{4} = 13$
 $\therefore$ Length = 13 cm and breadth = 11 cm
 Hence, area of rectangle
 $= l \times b = 13 \times 11$
 $= 143 \text{ cm}^2$

3. (a) Length of the arc = $\frac{2\pi r\theta}{360^\circ}$
 $= \frac{2 \times 22 \times 42 \times 72}{7 \times 360} = 52.8 \text{ cm}$

4. (d) Area of an isosceles right triangle with side
 $'a' = \frac{1}{2} a^2 = 200 \text{ cm}^2$
 $\therefore$ Side = $a = 20 \text{ cm}$
 Hence, hypotenuse
 $= \sqrt{a^2 + a^2}$
 $= \sqrt{2}a = 20\sqrt{2} \text{ cm}$



5. (b) Let the width of the walk be ' a ' m.



Given, length of outer rectangle = 20 m
 and breadth of outer rectangle = 10 m
 $\therefore$ Width of walk = a m
 $\therefore$ Length of inner rectangle
 $= (20 - a - a) = (20 - 2a)$ m
 and breadth of inner rectangle
 $= (10 - a - a) = (10 - 2a)$ m
 Now, area of inner rectangle
 $= (20 - 2a)(10 - 2a)$
 $\therefore (20 - 2a)(10 - 2a) = 96$ [given]
 $\Rightarrow 4a^2 - 60a + 104 = 0$
 $\Rightarrow a^2 - 15a + 26 = 0$
 $\Rightarrow (a - 13)(a - 2) = 0$
 But $a \neq 13$, so $a = 2$ m

6. (c) Side of the greatest square tile
 $=$ GCM of the length and breadth of the room
 $=$ GCM of 10.5 and 3 is 1.5 m
 Area of room = $10.5 \times 3 \text{ m}^2$
 $\therefore$ Number of tiles needed
 $= \frac{l \times b}{(\text{H.C.F. of } l \text{ \& } b)^2}$
 $= \frac{10.5 \times 3}{2.25} = 14 \text{ tiles}$

7. (c) Let side of a square be a . Area of square = a^2

$\therefore$ The side of a square be increased by 50%, then new side = $a + \frac{a}{2} = \frac{3a}{2}$

New area = $\frac{9a^2}{4}$

Increase in area = $\frac{9a^2}{4} - a^2 = \frac{5a^2}{4}$

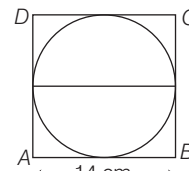
$\therefore$ Percent increase in area = $\frac{5a^2}{4a^2} \times 100$
 $= 125\%$

Shortcut Method

Side of square is increased by 50%.

$\therefore$ Net effect on area
 $= \left[50 + 50 + \frac{50 \times 50}{100} \right] \% = 125\%$

8. (d) Given, diameter of circle = Side of square = 14 cm
 $\therefore r = 7 \text{ cm}$



Area of circle = $\pi r^2 = \frac{22}{7} \times 7 \times 7$
 $= 154 \text{ cm}^2$

9. (c) Let original radius of circle be r .
 $\therefore$ Area of circle = πr^2
 $\therefore$ Radius of a circle is decreased by 50%, then
 New radius = $\frac{r}{2}$